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(SW)

PAGE
DATE

$$1) E(X) = 1.5$$

$$E(3X) = 4.5$$

$$2) \begin{aligned} P(X) &= K^2 - 48 \Rightarrow K^2 - 48 = 1 \Rightarrow K^2 = 49 \\ \sqrt{K^2} &= \sqrt{49} \quad K = 7 \end{aligned}$$

3) calculate $E(X - E(X))$

$$\begin{aligned} E(X - E(X)) &\Rightarrow E(X) - E(E(X)) \Rightarrow \\ E(X) - E(X) &= 0 \end{aligned}$$

$$4) \sum P(X) = 1 \Rightarrow 5K + 4K + 3K + 2K + K = 1 \Rightarrow 15K = 1$$

$$\cancel{K = \frac{1}{15}} \quad K = \frac{1}{15}$$

$$5) V(ax + by) = V(ax) + V(by) + 2\text{cov}(ax, by)$$

$$V(3X) + V(4Y) + 2\text{cov}(3X, 4Y)$$

$$\begin{aligned} &= 3^2 V(X) + 4^2 V(Y) + 0 = \\ &3^2 \cdot 0.2 + 0.5 \cdot 4^2 = 9.8 \end{aligned}$$